

Wow, you did it!

You connected your wallet, received the course token, and passed the token-gated access check.

What was built

For this FinTech exercise, a small token economy was created on the Sepolia testnet. The setup started with a simple ERC-20 course token, deployed as a custom token that can be held by student wallets and used by other smart contracts.

Next, a swap contract was added. This contract can hold the course token and exchange it against Sepolia ETH at a fixed rate. It also includes a redemption function, so users can send tokens back to the contract and receive Sepolia ETH in return, provided the contract has enough funds.

An NFT minting contract was then created. Students can mint a limited course NFT by paying with the custom ERC-20 token. The NFT points to metadata and an image hosted through Pinata/IPFS, while the contract enforces the maximum supply and the token-based minting price.

How the website works

A minimal GitHub-hosted website was prepared with three core functions: connect a MetaMask wallet, interact with the swap contract, and check whether the connected wallet holds enough course tokens.

The website does not need MetaMask to visually display the token in the asset list. Instead, it reads the token balance directly from the ERC-20 contract using `balanceOf(address)`. If the connected wallet holds at least 50 tokens, the website displays a success message and unlocks this PDF download.

Main takeaway

The important idea is that the blockchain state is the source of truth. Wallets, explorers, and websites are only interfaces. Even if a wallet interface fails to display a token, the smart contract balance still exists and can be used by the website, the swap contract, and the NFT contract.

This PDF is the token-gated reward: proof that your wallet passed the on-chain access condition.